Pre-A ST8/Q3LOCK OS-KMS, Ground, Gap, Counterterm, and Empty-Reference Route Split

TECT verification-first repository
claim `C6-SPACETIME-SIGNATURE`

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1 Purpose and scope

This note resumes T-054 from merged commit `7f0d5b44` and `EXP-000789`. It follows the requested order without identifying mathematically different objects:

1. phasewise periodic Osterwalder–Schrader/KMS reconstruction;
2. one common real-time dynamics;
3. distinct ground-state selection;
4. the broken-sector GNS gap;
5. an enlarged-counterterm continuum;
6. physical empty-space comparison.

Four subresults advance and four shortcut routes fail. Each positive-source sign phase reconstructs an abstract thermal system, but the two reconstructions do not yet supply one Hamiltonian-derived dynamics. The zero-temperature source energy has a strict cusp, but the resulting time-zero tangent states are not yet algebraic grounds. The symmetry-allowed local quartic basis is exactly nineteen-dimensional, but a regulator-removed continuum is not yet constructed. Finally, the same-Hamiltonian finite-volume variational identity freezes the earliest honest reference comparison, but it does not identify physical empty space.

All statements are T0 and claim-nonbearing. In particular, this note does not advance C6 or close CP1, Pre-A, A13, or physical Sector A.

2 Inherited fixed-lattice model

At one spatial site the configuration is $q_y \in \mathbb{R}^8$. The inherited zero-source finite-volume Hamiltonian is

$$H_\Lambda(0) = \sum_{y\in\Lambda}\left[-\frac{\hbar^2}{2\chi}\Delta_y + U_{Q3}(q_y)\right] + \frac{c}{2}\sum_{\langle yz\rangle}|q_y - q_z|^2,$$

with $g > 0$, $\lambda > 0$, $r < 0$, $c > 0$, and $\chi > 0$. The collective source is

$$H_\Lambda(h) = H_\Lambda(0) - hS_\Lambda, \qquad S_\Lambda = \sum_y Q_y, \qquad Q_y = \frac{1}{\sqrt{8}}\sum_{e=1}^{8} q_{y,e}.$$

The finite-temperature input from `EXP-000782` is the explicit sufficient regime

$$A_0 = \frac{8c\chi\theta_Q^2}{\hbar^2} > I_3, \qquad \theta_Q = \frac{-r}{3(g+\lambda)}, \qquad \beta > \beta_*.$$

It supplies two parity-related tempered Euclidean DLR measures $\mu_{\beta,+}$ and $\mu_{\beta,-}$. The ground-order input from `EXP-000789` is the beta-first dyadic sequence with $A_0 > J_3^2$, on which

$$\liminf_L \frac{\langle S_L^2 \rangle}{V^2} \geq \rho_* > 0, \qquad V = L^3.$$

No $a \to 0$ limit is hidden in these fixed-spacing premises.

3 Phasewise periodic OS/KMS reconstruction

Let T_s translate loop time and let Θ reflect the thermal circle. For a positive-half bounded local cylinder F, every finite-volume source law has the exact time-cut form

$$\int \overline{F(\Theta\omega)} F(\omega)\, d\mu_{L,h} = Z_{L,h}^{-1} \operatorname{Tr}(B_F^* B_F) \geq 0.$$

The integrand is bounded, continuous, and local. Temporal reflection positivity therefore survives first $L \to \infty$ and then $h \to 0^{\pm}$. Loop continuity similarly gives strong L^2 continuity of time translation.

The sharp-time algebra is fixed before either phase. Let $\mathcal{S}_{\text{conf}}$ be the norm closure of the unital star-algebra generated by finite-support rational-frequency characters

$$W_\xi(q) = \exp\left(i \sum_y \xi_y \cdot q_y \right), \qquad \xi_y \in \mathbb{Q}^8.$$

It is a separable unital commutative C-star algebra. The complete positive-temperature reconstruction interface is explicit:

- P1: path continuity and the product sup-norm bound give joint continuity;
- P2: the loop laws are time-translation invariant;
- P3: loop periodicity and commutativity give cyclic beta translation;
- P4: the time-cut form gives the positive reflection matrix;
- P-star: equal-time insertions multiply, the unit disappears, and every Green function is bounded by the product C-star norm.

The Birke–Froehlich reconstruction theorem therefore applies separately to each sign. It yields

$$(\mathcal{M}_{\beta,+}, \alpha^{\beta,+}, \omega_{\beta,+}), \qquad (\mathcal{M}_{\beta,-}, \alpha^{\beta,-}, \omega_{\beta,-}),$$

as abstract stochastically positive beta-KMS systems. Parity intertwines the two reconstructions. Finite first moments imply that a small rational-frequency collective character has different expectations in the two phases, so they remain distinguishable inside $\mathcal{S}_{\text{conf}}$.

This is phasewise, not common. The thermal generator is a Liouvillean, not a nonnegative vacuum Hamiltonian. Momentum and a full canonical Weyl algebra are also not obtained from this configuration reconstruction alone.

4 Common real-time dynamics remains open

The post-hoc direct sum

$$(\mathcal{M}_{\beta,+} \oplus \mathcal{M}_{\beta,-}, \alpha^{\beta,+} \oplus \alpha^{\beta,-})$$

exists for arbitrary unrelated systems and keeps the phase label as a central summand. It is not the thermodynamic limit of the exact zero-source local Hamiltonians on one state- and beta-independent labelled oscillator algebra. The route is therefore registered as

`NG-2026-08-09-PRE-A-ST8-Q3LOCK-POSTHOC-DIRECT-SUM-COMMON-DYNAMICS`.

The current theorem imports also miss the exact hypothesis class. The Weyl-integral anharmonicity used in arXiv:0909.2249 is bounded. Absorbing the quartic onsite term leaves the spatial term $-cq_y \cdot q_z$ unbounded. The audited subquadratic theorem does not include a cubic force. The resolvent algebra of arXiv:1605.05259 is the best common algebra candidate, but its available lattice theorem does not directly include this simultaneous quartic-force and unbounded-bilinear parent. This direct-import obstruction is

`NG-2026-08-09-PRE-A-ST8-Q3LOCK-CURRENT-COMMON-DYNAMICS-THEOREM-IMPORT-MISMATCH`.

The next constructive gate is to use smooth polynomial truncations on a fixed resolvent or energy-damped quasi-local algebra, prove truncation- and source-uniform energy-weighted Lieb–Robinson/Cauchy bounds, remove the truncation, and identify a common generator core.

5 Strict zero-temperature source cusp

Let $e_L(h) = E_{0,L}(h)/(8V)$. The locally uniform all-source limit $e(h)$ exists by `EXP-000780`. The `EXP-000789` broken doublet has

$$\langle S_L \rangle_+ = V m_L, \qquad m_L^2 = \frac{\langle S_L^2 \rangle}{V^2}, \qquad \epsilon_L \le \frac{\hbar^2}{4\chi V m_L^2}.$$

Using it as a trial vector at $h > 0$ gives

$$e_L(h) - e_L(0) \le \frac{\epsilon_L}{8V} - \frac{h m_L}{8}.$$

The excess term is $O(V^{-2})$, hence parity and the thermodynamic limit give

$$\boxed{e(h) - e(0) \le -\frac{|h|}{8}\sqrt{\rho_*}.}$$

Since e is even and concave,

$$D_+ e(0) \le -\frac{\sqrt{\rho_*}}{8}, \qquad D_- e(0) \ge +\frac{\sqrt{\rho_*}}{8}.$$

At differentiability sources tending to zero, Hellmann–Feynman, local compactness, and the uniform moment bounds select parity-related locally normal time-zero tangent candidates with opposite order at least $\sqrt{\rho_*}$. They become algebraic ground states only after a common α^h and convergence of its local generators to the zero-source core are proved.

6 The GNS gap is a separate gate

For a selected algebraic ground state define the state-vector Poincare gap

$$\Delta_\omega^P = \inf \sigma\Big(H_\omega|_{\Omega_\omega^\perp}\Big).$$

A positive value forces $\ker H_\omega = \mathbb{C}\Omega_\omega$. For a finite block B, locality gives

$$[S_B, [H, S_B]] = \frac{\hbar^2}{\chi}|B|$$

and the exact Rayleigh bound

$$\boxed{\Delta_\omega^P \le \frac{\hbar^2|B|}{2\chi\,\mathrm{Var}_\omega(S_B)}.}$$

Superlinear connected variance therefore forces the state-vector Poincare gap to vanish. Linear variance is only necessary: the full Poincare inequality must hold for every local core observable. A spectral gap defined above a degenerate zero eigenspace instead requires projection off the full kernel; ordinary variance does not decide it. Neither ground-vector simplicity nor a broken-sector GNS gap is claimed.

The collective classical Hessian has levels

$$K_\ell(k) = -2r + \lambda\frac{-r}{g}\ell + 2cE(k), \qquad \ell = 0, 2, 4, 6,$$

with positive minimum $-2r$. This removes an immediate tree-level Goldstone obstruction only; it is not a quantum gap proof.

7 Complete symmetry-allowed counterterm basis

The eight Q3 vertices are $\mathbb{F}_2^3$, and $\mathrm{Aut}(Q_3) = \mathbb{F}_2^3 \rtimes S_3$ has order 48. The 330 degree-four monomials split into exactly 19 orbits:

$$1\,O_4 + 3\,O_{31}^{(d)} + 3\,O_{22}^{(d)} + 6\,O_{211} + 6\,O_{1111}.$$

The bare vectors are

$$P_g = \frac{1}{4}O_4, \qquad P_\lambda = \frac{3}{4}O_4 - \frac{1}{2}O_{31}^{(1)} + \frac{1}{2}O_{22}^{(1)}.$$

For $B(P,Q) = \mathrm{tr}(P''Q'')$, exact rational expansion gives

$$\begin{aligned} B(W_4, W_4) = &\left(9g^2 + 54g\lambda + \frac{195}{2}\lambda^2\right) O_4 \\ &- (18g\lambda + 72\lambda^2)O_{31}^{(1)} + (12g\lambda + 71\lambda^2)O_{22}^{(1)} \\ &+ 18\lambda^2 O_{211}^{(1,1;2)} - 6\lambda^2 O_{211}^{(1,2;1)} + 4\lambda^2 O_{22}^{(2)}. \end{aligned}$$

Thus adding only the distance-two witness already misses two first-loop directions. Repeated basis-independent closure has ranks

$$2 \longrightarrow 4 \longrightarrow 9 \longrightarrow 19 \longrightarrow 19.$$

The partial repair is registered as

`NG-2026-08-09-PRE-A-ST8-Q3LOCK-PARTIAL-QUARTIC-`
`COUNTERTERM-ALL-SCALE-CLOSURE.`

The invariant quadratic matrices are the four Hamming-distance matrices A_0, A_1, A_2, A_3. A scalar term is separate, and temporal and spatial kinetic matrices must be allowed independently until Euclidean restoration is proved. This classifies the operator basis. It does not supply an $a \to 0$ flow, uniform coercivity, reflection-positive tightness, an OS continuum, or non-Gaussianity.

8 Physical empty-space comparison contract

At finite spatial volume and finite regulator,

$$\mathcal{F}_\beta(\sigma_{\emptyset,L}; H_L) - \mathcal{F}_\beta(\rho_{\beta,L}; H_L) = \beta^{-1} D(\sigma_{\emptyset,L} \| \rho_{\beta,L}) \geq 0,$$

and at zero temperature

$$\mathrm{Tr}(\sigma_{\emptyset,L} H_L) - E_{0,L} \geq 0.$$

Both are invariant under $H_L \mapsto H_L + cV\mathbf{1}$. A strict bulk claim requires a positive specific limit, not order-one finite-volume strictness.

Equilibrium KMS phases of one H, β have the same equilibrium free-energy density, and ground phases have the same ground-energy density. Another equilibrium phase cannot therefore be a strict empty comparator. Moreover, the symmetric `EXP-000789` ground already has long-range order, its broken doublets have nonnegative $O(1/V)$ total excess, and $q = 0$ is not a normalized quantum state. The rejected shortcut is

`NG-2026-08-09-PRE-A-ST8-Q3LOCK-EQUILIBRIUM-PHASE-AS-STRICT-EMPTY-REFERENCE.`

A future physical-empty branch must be preregistered as a constrained, metastable, or preparation state; use the same algebra, Hamiltonian, regulator, counterterms, geometry, stress tensor, units, and limit path; pass no-condensate and clustering tests; and have a positive specific scalar-shift-invariant gap.

9 Evidence map and next gate

The exact proof map is:

1. phasewise OS/KMS: closed in the abstract configuration-algebra scope;
2. common real-time dynamics: open after two registered shortcut failures;
3. zero-temperature source cusp and time-zero tangents: closed;
4. algebraic grounds and sector GNS gap: open;
5. full invariant counterterm basis: closed;
6. regulator-removed enlarged continuum: open;
7. finite-volume same-H reference identity: closed;
8. physical empty branch and positive specific comparison: open.

The next parallel ST8 gate is

`PA-CP1-ST8-Q3LOCK-RESOLVENT-ALGEBRA-EXACT-POLYNOMIAL-COMMON-ALPHA-CLOSURE.`

Only after it closes should the work identify the two phasewise systems as KMS states of one common alpha, promote the source tangents to algebraic grounds, and attempt the sector Poincare inequality. The cross-candidate T-054 admission freeze remains the overall primary priority.

10 Prior-art boundary

Periodic positive-temperature reconstruction, KMS theory, oscillator-lattice locality estimates on their stated domains, resolvent algebras, GNS Poincare criteria, multiscalar one-loop renormalization, and Gibbs relative entropy are prior mathematics. Primary route references are:

- Y. Kozitsky, *Equilibrium States, Phase Transitions and Dynamics in Quantum Anharmonic Crystals*, https://arxiv.org/abs/1806.08264;
- L. Birke and J. Froehlich, *KMS, etc.*, https://arxiv.org/abs/math-ph/0204023;
- B. Nachtergaele et al., *On the Existence of the Dynamics for Anharmonic Quantum Oscillator Systems*, https://arxiv.org/abs/0909.2249;
- D. Buchholz, *The Resolvent Algebra for Oscillating Lattice Systems*, https://arxiv.org/abs/1605.05259.

The repository-specific content is the exact Q3LOCK hypothesis map, the passage of the selected loop phases through the full periodic reconstruction interface, the `EXP-000780`/`EXP-000789` cusp composition, and the exact Q3 orbit/closure computation. No general-method novelty or historical-priority claim is made.

11 Devil's-advocate audit

1. **Objection: spatial reflection positivity was reused as temporal positivity. DISMISSED.** The time-cut trace factorization is a separate finite-volume identity and passes only bounded cylinders.
2. **Objection: reflection positivity alone gives a C-star KMS system. DISMISSED after repair.** The separable rational- character algebra and P1–P4/P-star map are declared explicitly.
3. **Objection: parity equivalence is already common dynamics. UPHELD as false.** It is an intertwiner of separately reconstructed phase-labelled systems.
4. **Objection: the source tangents are automatically algebraic grounds. UPHELD as too strong.** Common-alpha generator-core convergence is still required.
5. **Objection: the full finite-volume tunnelling gap decides the broken-sector GNS gap. UPHELD as false.** The state-vector Poincare inequality and connected variance are separate.
6. **Objection: ordinary variance also controls a gap above a degenerate zero eigenspace. UPHELD as false.** That notion needs the full-kernel projection; ground-vector simplicity remains open.
7. **Objection: the distance-two quartic alone repairs the continuum basis. DISMISSED.** Two new O_{211} terms occur at the same first loop, and exact closure reaches all nineteen directions.
8. **Objection: the nineteen-dimensional basis proves a continuum. UPHELD as false.** Bare flows, positivity, tightness, critical scaling, OS convergence, and non-Gaussianity remain open.

9. **Objection: the symmetric ground is physical empty space. DISMISSED.** It has extensive long-range order.

10. **Objection: a common scalar counterterm can reverse the reference sign. DISMISSED.** The relative identities are invariant under a common extensive scalar shift.

11. **Objection: this selects a Pre-A winner or advances C6. UPHELD as false.** The package is T0 and leaves both gates open.

12 Reproduction

Primary exact audit:

```
E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_cp1_st8_q3lock_os_dynamics_ground_gap_
counterterm_empty_route_split.py
```

Independent non-importing audit:

```
E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_cp1_st8_q3lock_os_dynamics_ground_gap_
counterterm_empty_route_split_independent.py
```

The primary implementation constructs the 48 cube automorphisms as $\mathbb{F}_2^3 \rtimes S_3$ and uses symbolic exact arithmetic. The independent implementation scans all 8! vertex permutations, uses a custom `Fraction` polynomial engine and its own row reduction, and imports neither the primary code nor its JSON. The expected child results are 88/88 and 120/120. The integrated verifier reruns both and checks all formal, freshness, generated-surface, scope, and unchanged-C6 firewalls.

13 Result footer

```
Result ID:               PA-CP1-ST8-Q3LOCK-PHASEWISE-OS-KMS-ZERO-T-
                         GROUND-CUSP-FULL-Q3-COUNTERTERM-AND-EMPTY-
                         REFERENCE-SPLIT.
Precise statement:       Phasewise beta-KMS; strict zero-T source cusp;
                         19D quartic and 4D quadratic basis; finite-volume
                         same-H reference contract; four shortcut failures.
Scope:                   T0 fixed-spacing sufficient regime; claim-nonbearing.
Dependencies:            EXP-000780--EXP-000782; EXP-000789--EXP-000790.
Evidence grade:          ANALYTIC + EXACT EXECUTED + INDEPENDENT + PDF QA.
Reproduction command:    python -X utf8 codes/foundations/
                         pre_a_cp1_st8_q3lock_os_dynamics_ground_gap_
                         counterterm_empty_route_split_verify.py
Expected output:         primary 88/88; independent 120/120; integrated
                         173/173; release checks PASS.
Falsification gate:      Any failed P1--P4/P-star input, cusp factor,
                         orbit/loop/closure coefficient, variance scope,
                         relative identity, parent ID, hash, or firewall.
Tier before / after:     C6 T1 / C6 T1; no claim or tier change.
No-overclaim statement:  Common alpha/KMS, algebraic grounds, GNS kernel and
                         gap, continuum, physical empty, C6, CP1, Pre-A,
                         A13, and physical Sector-A remain open.
Next required action:    RESOLVENT-ALGEBRA-EXACT-POLYNOMIAL-COMMON-ALPHA-
                         CLOSURE; keep the round-one admission freeze primary.
```